

Matias Freire

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EDUCATION

Stevens Institute Of Technology

Expected May 2028

B.S. in Computer Science

Hoboken, NJ

Awards: STEP Scholar: Selected from a pool of 200+ students based on outstanding leadership and commitment to the program.

Coursework: Algorithms, Data Structures, Computer Architecture and Organization, Systems Programming, Linear Algebra

Certifications: Bloomberg Market Concepts, Hugging Face AI Agents Fundamentals

TECHNICAL SKILLS

Programming: Java, Python, C/C++, JavaScript

Web Tools: TypeScript, React, Tailwind, Next.js

Developer Tools: Git, GitHub, VS Code, Jupyter Notebooks

Libraries: Dash, Pandas, NumPy, PyTorch, Scikit-learn, Streamlit

EXPERIENCE

Incoming Summer Software Engineering Intern

May 2026 – Aug 2026

Wabtec Corporation

Erie, PA

- Will work across embedded systems and modern software engineering, including C/C++ on Linux with Docker, as well as React, PyTorch, and AI development tools.

Deep Learning Research Assistant/Intern

Aug 2025 – Present

Advanced Quantum Materials Lab

Hoboken, NJ

- Researching preprocessing, synthetic data generation, and deep learning models for MoS defect characterization
- Researched ERM literature, built ERM models in COMSOL, and contributed to the first in its kind ML-driven optimization effort targeting shunt metal performance for device sensitivity.
- Designing, building, and maintaining a Dash-Plotly platform integrating open-source defect analysis pipelines

Machine Learning Ignite Fellow

Aug 2025–Jan 2026

AI4ALL

Remote

- Completed an applied machine learning fellowship implementing linear and tree-based models with pandas and scikit-learn
- Performed data pre-processing, feature engineering, and exploratory analysis on structured datasets as part of the programs Assignments

PROJECTS

MoS2 Image Synthesis & Analysis Platform | *PyTorch, Dash-Plotly, CSS*

[Github-Repo](#)

- Designed and built from scratch an end-to-end Dash-Plotly research platform for deep learning-based MoS defect analysis
- Automated synthetic image generation with configurable defects and realistic noise injection, integrating a ResUNet model for vacancy and polymorph classification

Colorstack Stevens Chapter Website | *Next.js, React, Tailwind, Typescript*

[Github-Repo](#)

- Built from scratch a modular, responsive Next.js and React website for the ColorStack Stevens chapter serving 60+ members
- Established a reusable UI system through rapid HTML/CSS prototyping and full implementation in TypeScript and Tailwind CSS

Interactive-Neural-Network-Digit-Classifer | *Python, Pytorch, Pygame, Streamlit*

[Github-Repo](#)

- Developed a Streamlit application for real-time handwritten digit recognition using a neural network achieving 96% test accuracy
- Integrated a Pygame-based drawing canvas enabling users to input custom digits for live model inference

LEADERSHIP & ACTIVITIES

Head of Extern Relations & Co-Chair of Technology | *Colorstack Stevens Chapter*

April 2024 – Present

- Led growth and technical engagement initiatives, increasing active membership by 50% and online engagement by 25% through social media strategy and 14+ events per semester
- Organized peer-led technical sessions, guiding members through 32+ data structures problems

STEP Tutor-Counselor/Resident Assistant (TCRA) | *Bridge Summer Program*

Jun 2025 – Aug 2025

- Led Calculus I/II and Physics recitations for 52+ incoming freshmen, strengthening STEM readiness through structured reviews and weekly after-hours tutoring for 25+ students
- Served as a Resident Assistant, personally mentoring and tutoring 52+ residents while enforcing residential policies and supporting academic success, wellness, and community standards